

Service Community Module

Development & coding plan for a geospatial patient-modeling + service-population module — V8-first, with a V9 cloud-migration leg.

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1. Overview & Principles

1.1 What we are building

The **Service Community module** lets an instructor model the *real clinical community* a cohort trains in — its **condition prevalence** (what illnesses/acuity actually occur locally) and its **demographic + visual makeup** (who the patients are, so avatars resemble the community) — and inject that context into MedSim simulations. It produces a portable, signed **community-block** consumed by V8 today and V9 later.

The module has two data planes that the research defines precisely and this plan wires together:

- **Clinical plane** — geospatial patient modeling (GPM) → locally prevalent conditions → scenario/case weighting.
- **Visual plane** — demographically-matched **synthetic** face cohorts (via FACE ENGINE) → avatars that look like the community.

Both planes are wrapped in the same governance envelope: a per-dataset **Sanitization & Data Declaration**, a per-element **Provenance / Source Declaration**, and a **standard community-block** exchange format.

1.2 Core principles

1. **Aggregate-only, no-PII by construction.** The community-block schema **cannot carry patient-level rows** — it is population statistics, public/aggregate sources, and synthetic targets only. `sanitization.contains_pii` is a hard `const:false` invariant; import fails closed on anything else. This is enforced at the schema layer, not by policy alone.
2. **Honest data, always labeled.** Every value declares its epistemic nature via a single load-bearing `source_type` enum: `factual-public | factual-nonpublic | hypothetical | expert-estimate`. Gap-fill is *allowed and expected*, but a filled gap is never dressed up as a source. Hypothetical data **must never claim a DOI** (false-provenance guard); synthetic faces always carry the visible "AI-GENERATED" label + C2PA marking inherited from FACE ENGINE.
3. **Provenance is a first-class product, not an afterthought.** Every element and every block can produce a machine-checkable reference list on demand (schema.org/Dataset JSON-LD + PROV-JSON), gated by a validator that fails on un-cited factual claims, dangling lineage, or missing stewards.
4. **Declare before you use.** No dataset enters a block without a completed Sanitization Declaration that satisfies **US (HIPAA), EU (GDPR), and India (DPDP 2023) simultaneously**, with a clear split between **[SELF]** (trained-instructor-attestable) and **[LEGAL]** (counsel/DPO/expert-required) fields.
5. **Robust, stable, easy to navigate.** Standalone service (independent deploy/restart, matches FACE ENGINE ops pattern), a linear declaration wizard, a gallery for reuse, and a versioned schema so blocks made today still import next year.
6. **Build on what exists.** GPM/M16, the demographic matrix (M1), the Condition Editor (151 presets/11 groups), the Community tab, `/api/geo/*`, the V8 export bridge, the C2PA/watermark/audit stack — all reused. The **new** work is: the condition-prevalence layer, the declaration/provenance capture, the block builder + export/import + signing, the instructor UI, and the V8 adapter.
7. **Standalone-first, contract-coupled.** V8 (and later V9) couple to the module through a **versioned contract + per-product adapter** — never internals — mirroring the identity-keystone pattern already used across the ecosystem.

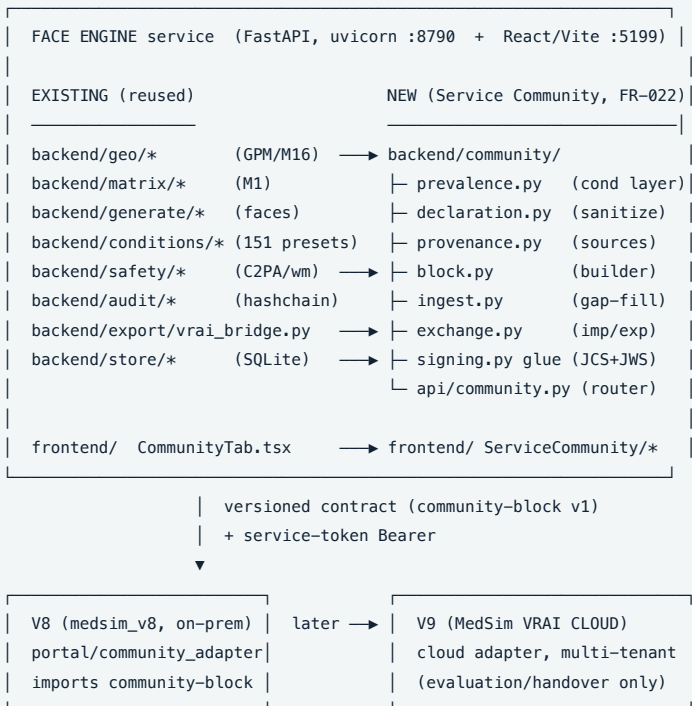
1.3 Non-goals (v1)

- No patient-level data ingestion, ever.
- No generation of net-new clinical case *content* — v1 does **prevalence-weighted selection** from the existing case library (new-case generation deferred).
- No production DataCite DOI minting in v1 (internal persistent URIs `tbsim:...`; DOI minting is an open decision, §10).
- V9 cloud is **evaluation + handover only** in this plan; it is not implemented here.

2. Architecture

2.1 Shape: standalone service alongside V8, reusing the FACE ENGINE service pattern

The Service Community module is delivered as a **module inside the existing FACE ENGINE repo/service** (`github.com/medsim-vrai/face-engine`), not a separate deployable. Rationale: the two data planes it needs — GPM and face-cohort generation — already live there, so co-locating avoids a network hop for the module's core loop and reuses the service's audit/RBAC/C2PA stack verbatim. It is exposed under a **new FastAPI router** and a **new frontend tab**, and it stands alongside V8 exactly as FACE ENGINE already does.



Run (unchanged from FACE ENGINE): `cd repo && .venv/bin/activate && uvicorn backend.app:app --port 8790 + cd frontend && npm run dev`.
New router registered in `backend/app.py` alongside the existing `geo`, `generate`, `conditions`, `vrai` routers.

2.2 How V8 links to it

V8 is coupled through a **thin adapter + a versioned contract**, never FACE ENGINE internals:

- **Contract:** the `community-block/1.0.0` JSON Schema (§3) is the sole interface. V8 validates any block it imports against the schema it ships.
- **Adapter (new, V8-side):** `medsim_v8/portal/community_adapter.py` — pulls or receives a community-block, validates it, verifies its `integrity` (JCS re-hash + optional JWS), rejects on `contains_pii ≠ false` or unknown-major `spec_version`, then projects the block onto V8's existing surfaces:
- `clinical_prevalence[]` → scenario/case **weighting** in the authored-content / scenario layer (`authored_content.py`, scenario selection).
- `demographics` + `face_cohort_spec` → a face-cohort request to FACE ENGINE via the **already-built** `request_face_from_engine()` path (`vrai_faces.py`, `POST /portal/face/request-engine/...`) and the M15 control adapter (`POST /api/vrai/request` → drop-file into V8).
- block metadata + provenance → a read-only "Community context" panel in the V8 portal.
- **Transport:** two supported modes — (a) **file exchange** (educator exports a `.tbc.b.json` block, imports into V8; zero network), and (b) **service-to-service** pull from FACE ENGINE with a **service-token Bearer** on the contract route (consistent with the ecosystem's contract-route auth). File exchange is the v1 default (robust, offline-friendly, matches how educators actually share).

Sequencing guard: the V8 adapter is built and validated against fixtures **before** any live V8↔service wiring is switched on. Link only after the block round-trips cleanly in isolation (P4).

2.3 The V9 cloud variation (later leg)

V9 gets its **own adapter**, not a fork of the module. Because the block is transport-agnostic and residency-neutral (aggregate/public/synthetic only), the same `community-block/1.0.0` contract carries into V9 unchanged. What changes for cloud is *around* the block, not the block: multi-tenant scoping, object-store residency, per-tenant signing keys, and the contract-hub coupling (identity keystone). Detailed in §11.

3. The Community-Block Data Model

The single portable artifact is the **community-block**: an aggregate-only, versioned JSON document consumed by V8 and V9 and shareable between educators. It embeds the demographic mix, condition prevalence, face-cohort spec, the sanitization declaration, and the provenance/source list. **Invariant: no patient-level rows, ever.**

3.1 Schema location & versioning

- Canonical schema: `backend/community/schemas/community-block/1.0.0.json` (JSON Schema Draft 2020-12), `$id`: `https://tbsim.company/schemas/community-block/1.0.0.json`, `format`: `"tbsim.community-block"`.
- **Two independent SemVer axes:**

- `spec_version` — the schema. Consumers **route on major**: accept same-major, reject unknown-major, tolerate unknown minor/patch (forward-compat: additive-only within a major; `additionalProperties:false` on core objects means new fields ship as a minor bump + relaxed sub-objects).
- `identity.block_version` — the *content* revision; `identity.block_id` (UUID) is stable across revisions of the same community; `identity.supersedes` links a revision to what it replaces.

3.2 Top-level structure (required unless noted)

Key	Purpose	Standard anchor
<code>format / spec_version</code>	Identity + schema routing	SemVer
<code>identity</code>	<code>block_id</code> , <code>block_version</code> , <code>name</code> , <code>created_at</code> , <code>created_by</code> (org/role, no PII), <code>supersedes</code> , <code>tags</code>	—
<code>geography</code>	<code>scope</code> (nation...zcta/hsa/custom), <code>label</code> , <code>gazetteer</code> (FIPS/GEOID/ISO), <code>boundary</code>	GeoJSON RFC 7946 , WGS84/CRS84 only, no crs member
<code>demographics</code>	<code>population_estimate</code> (count, not a member list), <code>as_of</code> , <code>strata[]</code>	FHIR MeasureReport <code>stratifier.stratum</code> shape
<code>clinical_prevalence[]</code> (opt)	per condition: <code>condition</code> (CodeableConcept), <code>measure</code> (<code>prevalence_fraction</code> / <code>rate_per_100k</code> / <code>incidence_per_100k</code> + CI), optional <code>stratified_by[]</code> , <code>source_ref[]</code>	OMOP vocab pattern / FHIR CodeableConcept
<code>sdoh[]</code> (opt)	aggregate SDOH indices (SVI/ADI/uninsured %) — indices only	FHIR/gravity coding
<code>face_cohort_spec</code> (opt)	<code>cohort_size</code> , <code>targets[]</code> (axis/value/fraction), <code>derive_from_demographics</code> , <code>synthetic_only:const true</code>	FACE ENGINE M1 axes
<code>sanitization</code>	the no-PII declaration (§3.4) — required	§7 declaration
<code>provenance</code>	source list + per-element declarations (§3.5) — required	schema.org/Dataset + PROV-O + DataCite
<code>meta</code> (opt)	schema.org/Dataset projection for catalog/discovery	schema.org/Dataset
<code>integrity</code> (opt but required for signed export)	<code>canonicalization:"RFC8785"</code> , <code>hash_alg</code> , <code>content_hash</code> , optional detached <code>signature</code> (JWS)	RFC 8785 (JCS) + RFC 7515

Shared `$defs`: `CodeableConcept` (`system / code / display`), `Stratum` (`axis / value / fraction|count`, `count ≥ min_cell_size` or null), `GeoJSON` (`Feature / FeatureCollection`, `not:{required:[crs]}`).

3.3 Two embedded declarations, co-located and mutually referential

Every block carries **two orthogonal declarations** that answer different questions and stay consistent by construction:

- Sanitization** — “*Is this safe to expose?*” (it is a `PROV Activity` — used `raw`, `generated` exposed).
- Provenance** — “*Where did this come from, can I cite it?*” (the `PROV Entity` + its `wasDerivedFrom` / `hadPrimarySource` edges).

They cross-reference (`SourceDeclaration.sanitization_ref` ↔ `SanitizationDeclaration`), roll up into the same block manifest, and — critically — when sanitization substitutes a synthetic value for a real one, the element's `source_type` **flips to** `hypothetical / expert-estimate` and the chain records the substitution, so the two can never silently disagree.

3.4 Embedded sanitization block (block-level roll-up)

```
"sanitization": {
  "contains_pii": false,          // const:false – import MUST fail closed otherwise
  "aggregate_only": true,         // const:true – no patient-level rows
  "min_cell_size": 11,           // HIPAA-style k≥11 small-cell suppression
  "suppressed": true,
  "method": "small-cell-suppression+rounding",
  "synthetic_faces": true,
  "declared_at": "...", "declared_by": "role:clinical-lead",
  "declaration_ref": "decl:..." // → full cross-regime Declaration record (§7)
}
```

The block-level `sanitization` is the machine-enforced invariant; the full US/EU/India **Declaration** (§7, Sections A–G) is stored as a linked record and referenced by id, so the block stays small while the audit trail stays complete.

3.5 Embedded provenance (`sources[]` + per-element `SourceDeclaration`)

```
"provenance": {
  "sources": [{
    "id": "src-cdc-places-2024", "name": "CDC PLACES", "publisher": "CDC",
    "url": "https://...", "vintage": "2024", "license": "US-PD",
    "access": "aggregate-public", "retrieved_at": "2026-06-30T..."
  }],
  "generator": "gpm@1.4.0 / face-engine@0.9.0"
}
```

Each **stat** in `clinical_prevalence / sdoh / demographics` references a source via `source_ref[]`. Under the hood each referenced element carries a full `SourceDeclaration` (`id`, `source_name`, `source_type`, `citation`, `url/doi`, `retrieval_date`, `confidence` `high|medium|low`, `is_estimate`, `steward`, `derived_from[]`, `license`, `sanitization_ref`). `source_type` deterministically drives the PROV relation, the C2PA `digitalSourceType`, and the citation obligation:

source_type	PROV relation	citation obligation	
factual-public	hadPrimarySource	MUST have `url\`	doi + retrieval_date`
factual-nonpublic	wasDerivedFrom + wasAttributedTo	MUST have steward + access note	
expert-estimate	wasAttributedTo (the SME/panel)	MUST name expert + is_estimate=true	
hypothetical	(no source relation) + com.tbsim.synthetic{real_person:false}	is_estimate=true ; MUST NOT carry a doi	

3.6 Machine-checkable reference-list export (one source of truth, three serializations)

- (a) **schema.org/Dataset JSON-LD** — web-discoverable; hasPart per element, isBasedOn lineage, citation, license, identifier.
- (b) **PROV-JSON graph** — verifiable lineage; each SourceDeclaration → prov:Entity, steward → prov:Agent via wasAttributedTo, retrieval → prov:Activity.
- (c) **C2PA carrier** — when a block/element is exported with an asset (e.g., the face cohort), the same facts ride inside a com.tbsim.provenance custom assertion beside FACE ENGINE's existing com.tbsim.synthetic + c2pa.actions, with sources as C2PA ingredients — cryptographically bound and tamper-evident.

3.7 Types

The schema maps 1:1 to **pydantic v2** models (`backend/community/models.py`) for the service and to **TypeScript** interfaces (`frontend/ServiceCommunity/types.ts`) for the UI, generated from the same JSON Schema so drift is impossible.

4. Modules / Components

All new code lives under `backend/community/` (Python 3.14, FastAPI, pydantic v2) and `frontend/ServiceCommunity/` (React/Vite/TS), reusing existing service modules.

4.1 Ingestion + gap-fill with labels — `backend/community/ingest.py`

- **Inputs:** (a) GPM auto-pull (CDC PLACES `eav7-hnsx`, ACS via existing `geo/live.py`), (b) educator CSV/JSON upload of aggregate stats, (c) manual entry for gaps.
- **Gap-fill with mandatory labels:** any value not sourced from a public dataset must be entered as `hypothetical` or `expert-estimate`, and the UI **forces** `is_estimate=true` + `confidence` + a steward. This is how factual-nonpublic and hypothetical inputs fill local-population gaps *without* laundering their epistemic status.
- **Output:** normalized candidate elements, each a draft `SourceDeclaration`, staged for the block builder.
- **Endpoints:** `POST /api/community/ingest` (upload), `POST /api/community/ingest/manual`.

4.2 Declaration / provenance capture — `backend/community/declaration.py`, `backend/community/provenance.py`

- `declaration.py` — the cross-regime Sanitization Declaration (§7) as a pydantic model with Sections A–G, per-field [SELF] / [LEGAL] tagging, and a **gate**: if any [LEGAL] field is invoked, `legal_review_status` is required before the declaration can back an export.
- `provenance.py` — builds `SourceDeclaration` + `ProvenanceManifest`, computes `provenance_summary` (counts by `source_type`) and `weakest_link` (`min(confidence)` + un-cited factual elements), and emits the three serializations (§3.6). Reuses FACE ENGINE's `safety/provenance.py` conventions (`com.tbsim.*`, `confidence`, `is_estimate`, `ingredients`).
- **Validator** (`provenance.validate_block`) — hard-fails export if: a `factual-public` element lacks `url|doi+retrieval_date`; a `hypothetical` element carries a `doi`; any element lacks a `steward`; any `derived_from` is dangling; the block lacks the 6 DataCite mandatory properties; `sanitization.contains_pii` ≠ `false`; or any [LEGAL] field lacks legal sign-off. (Gate-vs-report is an open decision, §10 — default: **gate**.)

4.3 Geospatial profiler reuse — backend/community/prevalence.py

- Calls existing GPM: `geo/pipeline.resolve()` + `geo/profile.build_profile()` (place → `GeoResolution` + `CommunityProfile` with census/BRFSS/WHO strata). **No new geo code.**
- **New:** the condition-**prevalence layer** — maps resolved demographics → locally prevalent conditions using CDC PLACES + published prevalence tables, emitting `clinical_prevalence[]` entries with `measure`, `CI`, and `source_ref`. Ships a `prevalence_sources.json` reference set (public/aggregate only). Where local data is missing, prevalence entries are labeled `expert-estimate` / `hypothetical` per §4.1.

4.4 Block builder — backend/community/block.py

- Assembles `geography` (GeoJSON from GPM), `demographics` (strata), `clinical_prevalence`, `sdoh`, `face_cohort_spec` (derived from demographics or explicit targets), plus the embedded `sanitization` + `provenance`.
- Validates against the JSON Schema, runs the provenance validator, and stamps `meta`.
- **Endpoints:** `POST /api/community/blocks` (build/save), `GET /api/community/blocks`, `GET /api/community/blocks/{id}`, `POST /api/community/blocks/{id}/revision`. Persisted in the existing SQLite store (`backend/store/db.py`, new `community_block` table).

4.5 Export / import with signing — backend/community/exchange.py, signing.py glue

- **Export:** canonicalize block minus `integrity` per **RFC 8785 (JCS)** → SHA-256 → `content_hash` → optional detached **JWS** (EdDSA/ES256, reusing FACE ENGINE `certs/` dev chain) → write `integrity`. Emits `<name>.tbcb.json` + the reference-list serializations; when bundled with a face cohort, embeds `com.tbsim.provenance` into the C2PA manifest.
- **Import:** parse → schema-validate → route on `spec_version` major → verify `integrity` (JCS re-hash + JWS) → **fail closed** on `contains_pii ≠ false`, unknown major, dangling lineage, or broken hash.
- **Endpoints:** `POST /api/community/blocks/{id}/export`, `POST /api/community/import`.

4.6 V8 adapter — medsim_v8/portal/community_adapter.py (V8-side, new, uncommitted per pattern)

- Validates + verifies an imported block, then projects it: prevalence → scenario weighting; demographics/face_cohort → face request via existing `request_face_from_engine()` / M15 control adapter; provenance/meta → read-only portal panel.
- Guards: no-overwrite on existing portraits (already present), reject on schema/PII/integrity failure. Gated behind a config flag until P4 validation passes.

5. UI — Instructor "Service Community" area

New top-level tab `frontend/ServiceCommunity/` (React/Vite/TS), sibling to the existing `CommunityTab.tsx`, whose UX it ports and extends. Design goals: **linear, low-cognitive-load, hard to get wrong, upgradeable.**

5.1 Screens

1. **Gallery / home** — cards of saved community-blocks (name, place, population, `provenance_summary` badge, sanitization status, version). Actions: New, Open, Duplicate→revision, **Export**, **Import**. This is the "easy navigation" anchor — everything starts here.
2. **Define Service Area** — place search → GPM resolve → map (GeoJSON) + resolved demographic profile. One field, one result.
3. **Develop Context** — two panels: **Demographics** (editable strata) and **Condition Prevalence** (prevalence table, add/adjust, each row shows its `source_type` chip and confidence). Gap rows are visually distinct and force a label.
4. **Declaration flow** (the wizard, §5.2).
5. **Face Cohort** — derive targets from demographics or set explicit; trigger FACE ENGINE cohort; select faces (reuses Community tab picker + AI badge).
6. **Review & Export** — provenance summary, `weakest_link`, validator result (pass/fail with reasons), sign + export `.tbcb.json`.

5.2 The declaration flow (Sanitization + Provenance)

A **stepper** that mirrors research Sections A–G, one section per step, with inline `[SELF]` / `[LEGAL]` badges:

- **A** Identity & provenance → **B** Data nature/source/granularity (drives which later fields are mandatory) → **C** De-id method (Safe Harbor 18-item checklist is a literal checklist; Expert Determination flagged `[LEGAL]`) → **D** Re-id risk + small-cell check → **E** three jurisdiction attestations (US/EU/India, always all three) → **F** legal basis/consent (only if personal in any regime) → **G** sign-off.
- **Progressive disclosure:** hypothetical/aggregate paths collapse to a short path; individual-level factual data expands the full de-id + legal branch.
- **Legal gate:** any `[LEGAL]` field toggles a "Legal review required" banner and blocks final export until `legal_review_status` is filled.
- **Reusable:** a completed declaration is saved and can be attached to future blocks (dedup by dataset).

5.3 Navigation, robustness, upgradeability

- Persistent left nav (Gallery · Define · Develop · Declare · Faces · Review); breadcrumb; autosave drafts to the store (no lost work).
- Every destructive action confirmed; every block immutable once exported (new edits = new revision via `supersedes`).
- **Upgradeability:** the UI reads `spec_version` and renders a "schema vN" badge; unknown-minor fields are preserved on round-trip (never dropped), so newer blocks survive an older UI.

6. Reuse Map (build-on vs new)

Capability	Home / file	Status	Action
Geo resolve/profile (GPM/M16)	<code>backend/geo/{pipeline,profile,live,model}.py , /api/geo/{resolve,profile}</code>	built	reuse as-is
Demographic strata (census/BRFSS/WHO)	<code>population_stratification_schema.md , backend/matrix/*</code>	built	reuse — block strata mirror it
Cohort face generation	<code>backend/generate/* , POST /api/geo/generate , /cohort</code>	built	reuse for <code>face_cohort_spec</code>
Condition presets (151/11 groups)	<code>backend/conditions/presets.py , /api/conditions</code>	built	reuse for on-face conditions
Community operator tab	<code>frontend/.../CommunityTab.tsx</code>	built	port + extend into ServiceCommunity
C2PA + watermark + <code>com.tbsim.*</code>	<code>backend/safety/{provenance,signing}.py</code>	built	reuse — provenance/sanitization assertions ride the same manifest
Hash-chain audit + RBAC	<code>backend/audit/* , OPERATOR_TOKENS</code>	built	reuse for declaration/export audit
SQLite store (FaceRecord)	<code>backend/store/{db,models}.py</code>	built	extend — add <code>community_block , declaration</code> tables
V8 export / drop-file / M15	<code>backend/export/vrai_bridge.py , /api/vrai/request</code>	built	reuse for face delivery
V8 face request button	<code>medsim_v8/.../vrai_faces.py , POST /portal/face/request-engine/{cid}</code>	built (uncommitted)	reuse from adapter
Condition-prevalence layer	<code>backend/community/prevalence.py</code>	NEW	build
Sanitization/Provenance capture	<code>backend/community/{declaration,provenance}.py</code>	NEW	build
Block builder + schema	<code>backend/community/{block.py,schemas/...}</code>	NEW	build
Export/import + signing (JCS/JWS)	<code>backend/community/{exchange,signing}.py</code>	NEW	build
Instructor UI	<code>frontend/ServiceCommunity/*</code>	NEW	build
V8 adapter	<code>medsim_v8/portal/community_adapter.py</code>	NEW	build

7. Governance & Compliance

7.1 Cross-regime declaration (US / EU / India — one record satisfies all three)

The declaration captures data **nature** (factual-public / factual-nonpublic / hypothetical), **granularity** (individual vs aggregate/community), **de-id method**, a **re-identification-risk statement**, and **three independent jurisdiction attestations** — because a dataset can be out-of-scope under HIPAA yet regulated under GDPR or DPDP. Key rules baked in:

- **HIPAA:** Safe Harbor (18 identifiers removed + actual-knowledge) is **[SELF]**; Expert Determination is **[LEGAL]**. Aggregate data is often outside PHI but still needs a stated basis + small-cell check.
- **GDPR:** anonymisation removes data from scope (Recital 26) but the bar is high (defeat singling-out/linkability/inference) and any "anonymous" reliance is **[LEGAL]**; **pseudonymisation is still personal data** (EDPB 01/2025) — never an out-of-scope claim.
- **DPDP 2023: no de-id safe harbor, no sensitive-data tier** — almost every scope/exemption claim is **[LEGAL]**; conservative default = treat as personal + document a lawful ground; the s.17 research exemption's safeguards were not fully notified as of 2026 (re-check before go-live).

[SELF] vs [LEGAL] split (keeps the module usable without over-promising): **[SELF]** = data-nature/source/granularity classification, the HIPAA 18-item checklist, actual-knowledge for Safe Harbor, small-cell check, steward sign-off. **[LEGAL]** = any anonymisation/Expert-Determination *claim*, GDPR Recital-26 reliance, all DPDP scope/exemption conclusions, every lawful-basis/consent field, and use of not-yet-public factual data.

7.2 Audit, residency, AI-labeling

- **Audit:** every declaration, block build, export, and import is written to the existing **hash-chained audit log** (`backend/audit/log.py , verify_chain` detects edits/deletes). Declarations are time-bound (re-review-due date), since re-identification risk drifts.
- **Residency:** v1 is on-prem/standalone; blocks are aggregate/public/synthetic only, so they carry no residency burden. V9 adds per-tenant residency (§11).
- **AI-labeling:** synthetic faces carry the visible "AI-GENERATED" label + DWT-DCT watermark + C2PA `com.tbsim.synthetic{real_person:false}` inherited from FACE ENGINE; the block's `face_cohort_spec.synthetic_only:const true` asserts no real portraits.

8. Phased Plan with Milestones

Each phase lists concrete coding tasks, deliverables, and tests. All work is additive; commit per milestone.

P0 — Scaffold & schema (foundation)

- **Tasks:** create `backend/community/` package + `api/community.py` router (registered in `app.py`); author `schemas/community-block/1.0.0.json`; generate pydantic models (`models.py`) + TS types; add `community_block` + `declaration` tables to the store.
- **Deliverables:** schema file, models, router with `/api/health` extension reporting `community: enabled`.
- **Tests:** schema validates the canonical example + rejects a `contains_pii:true` block, a `crs`-bearing GeoJSON, and an unknown-major block; pydantic round-trips the example.

P1 — Prevalence layer + ingestion + gap-fill

- **Tasks:** `prevalence.py` (GPM → conditions, `prevalence_sources.json`); `ingest.py` (CSV/JSON upload + manual entry with forced labels); wire `source_type` gap-fill rules.
- **Deliverables:** `POST /api/community/ingest`, `/ingest/manual`; prevalence for a fixture place (e.g., Hartford, already live in geo).
- **Tests:** prevalence produced for a fixture; manual gap entry forces `is_estimate` + steward; upload rejected if a factual row lacks a source.

P2 — Declaration + provenance + validator

- **Tasks:** `declaration.py` (Sections A–G, [SELF]/[LEGAL] tagging, legal gate); `provenance.py` (SourceDeclaration, manifest, three serializations, validator).
- **Deliverables:** `POST /api/community/declarations`; reference-list export (JSON-LD + PROV-JSON).
- **Tests:** validator fails on un-cited factual / hypothetical-with-DOI / dangling lineage / missing steward / missing DataCite props / unsigned [LEGAL]; validator passes a clean block; JSON-LD + PROV-JSON validate.

P3 — Block builder + export/import + signing

- **Tasks:** `block.py` (assemble/validate/persist/revision); `exchange.py` (JCS canonicalize, SHA-256, JWS sign, export/import); C2PA `com.tbim.provenance` glue.
- **Deliverables:** build/export/import endpoints; `.tbc.b.json` round-trips; signed + verified.
- **Tests:** JCS canonicalization is stable (byte-identical across key reorder); hash + JWS verify; import fails closed on tamper/PII/unknown-major; block→export→import→block is lossless (incl. unknown-minor field preservation).

P4 — V8 link (the alongside→link step)

- **Tasks:** `community_adapter.py` (V8-side): validate/verify → project prevalence to scenario weighting + demographics/face_cohort to face request + provenance panel; config flag; face delivery via existing bridge.
- **Deliverables:** an imported block drives V8 scenario weighting + a matched face cohort + a read-only context panel, verified live on one community.
- **Tests:** adapter rejects bad blocks; end-to-end file-exchange import in V8; face cohort delivered via `request_face_from_engine`; no-overwrite guard holds.

P5 — UI

- **Tasks:** `frontend/ServiceCommunity/*` — Gallery, Define, Develop, Declaration stepper, Face Cohort, Review/Export; port Community tab picker; autosave; schema-version badge.
- **Deliverables:** full instructor flow in browser, verified live.
- **Tests:** Playwright/RTL: new→define→develop→declare→faces→export→re-import happy path; legal gate blocks export; unknown-minor round-trip preserved in UI.

P6 — Validation

- **Tasks:** end-to-end validation on 2–3 real communities; adversarial review of the validator + import fail-closed paths + declaration legal gate (same 4-lens pattern used on the Condition Editor); operator docs.
- **Deliverables:** validation report; hardened fixes; instructor guide + FAQ (mirror FACE ENGINE `docs/support/`).
- **Tests:** full regression green; audit chain verifies across the whole flow.

P7 — V9 evaluation & handover (later leg)

- **Tasks:** §11 migration evaluation + handover checklist; **no V9 implementation** here.
- **Deliverables:** V9 migration memo + handover doc.

9. Testing & Validation Strategy

- **Schema conformance:** every fixture block validated against `1.0.0.json`; negative fixtures for each hard invariant (PII, `crs`, small-cell < k, unknown-major).
- **Property tests:** JCS canonicalization stability under key/whitespace permutation; hash/JWS verify; round-trip losslessness incl. forward-compat (unknown-minor preservation).
- **Provenance validator matrix:** one test per fail condition + a passing golden block; JSON-LD validated against schema.org, PROV-JSON against PROV-JSON schema.
- **Declaration logic:** [SELF]/[LEGAL] branching per data-nature/granularity path; legal gate blocks export.

- **Integration:** GPM reuse (real Hartford/live-census path), face-cohort delivery via existing bridge, V8 adapter projection.
- **Adversarial review (P6):** import fail-closed, false-provenance guard, concurrent block edits, empty/degenerate blocks — reusing the multi-lens review pattern proven on the Condition Editor.
- **Ops parity:** unit + `pytest` green (Python 3.14), `ruff` clean, `tsc` clean, matching the existing repo bar.

10. Risks & Open Decisions

1. **Identifier policy** — real DataCite DOIs (needs membership/allocator) vs internal `tbid:...` URLs. *Default: internal URLs for v1; DOI minting deferred.*
 2. **C2PA signing trust** — FACE ENGINE self-signs (untrusted). For third-party-provable provenance, a trusted/anchored credential + RFC-3161 TSA may be needed. *Default: structured-but-dev-signed for v1.*
 3. **Validator: gate vs report** — hard-block export on un-cited factual elements (strict FORCE11 Evidence) vs emit-with-warning + audit entry. *Default: gate; revisit with instructors.*
 4. **DPDP s.17 safeguards not fully notified (2026)** — the research exemption cannot be safely relied on until prescribed standards land; conservative default holds. **Re-check before go-live.**
 5. **Synthetic-derived-from-real ambiguity** — AI faces trained on real datasets sit in unsettled legal territory across all three regimes; flagged [LEGAL], position technique/jurisdiction-dependent.
 6. **UK divergence** — EDPB (strict) vs UK ICO ("motivated intruder"); if UK data/users are in scope, add a separate UK attestation rather than folding into "EU".
 7. **v1 clinical-conditions scope** — prevalence-weighted selection vs new-case generation. *Default: selection.*
 8. **Expert-estimate governance** — single SME vs Delphi panel; whether low-confidence estimates need a second steward. *Open.*
 9. **V8↔service transport** — file exchange (default) vs service-to-service pull with service-token. *Default: file-first, service optional.*
-

11. V9 Migration Evaluation (later leg)

11.1 What carries unchanged

- The **community-block contract** (`community-block/1.0.0`) — transport- and residency-neutral by design; the same schema, validator, and export/import serve V9.
- **Aggregate-only / no-PII invariant** — makes the block cloud-safe: no residency burden on the block itself.
- **Provenance + sanitization declarations**, the JCS/JWS integrity, and the reference-list export.
- The **module logic** (prevalence, block builder, exchange) — pure functions over the schema, portable as-is.

11.2 What changes for cloud / multi-tenant

- **Adapter, not fork:** V9 gets its own `community_adapter` coupling through the same contract via the **integration-hub / identity keystone**, not FACE ENGINE internals.
- **Multi-tenant scoping:** `block_id`/declarations become tenant-scoped; RBAC moves from `OPERATOR_TOKENS` to per-tenant roles under the keystone.
- **Residency & storage:** blocks move from SQLite/file to per-tenant object storage; signing keys become per-tenant (KMS), replacing the shared dev `certs/`.
- **Signing trust:** cloud is the moment to upgrade to a trusted/anchored C2PA credential + TSA (Risk #2).
- **Face delivery:** V9 avatar export target already exists in the bridge (`target:"v9"`); needs FACE ENGINE reachable cloud-side (Azure/AWS move, not GoDaddy) — a known deferred item.
- **Local-context tie-in:** V9's existing `/ui/local-context` should *consume* the block, not duplicate it.

11.3 Handover checklist

- ☐ `community-block/1.0.0` schema + validator packaged for V9 import (versioned, same-major).
 - ☐ V9 `community_adapter` spec against the contract (no internals).
 - ☐ Tenant-scoping + RBAC mapping to the keystone defined.
 - ☐ Residency + per-tenant KMS signing plan.
 - ☐ Trusted C2PA credential + TSA decision resolved (Risk #2).
 - ☐ FACE ENGINE cloud reachability decided (for `target:"v9"` face delivery).
 - ☐ DPDP s.17 / GDPR-EDPB status re-checked at V9 go-live (Risks #4, #6).
 - ☐ Local-context consumption (not duplication) confirmed.
 - ☐ Migration memo + audit-continuity plan (hash-chain across on-prem→cloud) delivered.
-

Key file paths (grounding):

- FACE ENGINE service: `/Users/petermarotta/Documents/Claude/Projects/MedSim VRAI-FACE GENERATOR/` — reuse `backend/geo/*`, `backend/matrix/*`, `backend/generate/*`, `backend/conditions/presets.py`, `backend/safety/{provenance,signing}.py`, `backend/audit/*`,

- ```
backend/store/*, backend/export/vrai_bridge.py, frontend/.../CommunityTab.tsx.
```
- New module: `backend/community/` + `backend/api/community.py` + `frontend/ServiceCommunity/` + `backend/community/schemas/community-block/1.0.0.json`.
  - V8 adapter: `/Users/petermarotta/Documents/Claude/Projects/Scenario` structure to support character engagement/`medsim_v8/portal/community_adapter.py` (reuse `vrai_faces.py`, `POST /portal/face/request-engine/{cid}`).
  - Concept/feature docs: `MedSim VRAI-FACE GENERATOR/docs/SERVICE_COMMUNITY_feature-concept.md`, `medsim_v8/vrai-faces/docs/FR-022-service-community.md`.

## Appendices — research artifacts

The concrete, citable artifacts produced by the research phase (web-grounded). These are the schemas and reference base the plan refers to; keep them versioned with the plan.

## Appendix A — Sanitization & Data Declaration schema (US · EU · India)

### "Sanitization & Data Declaration" Schema

A single declaration an instructor completes per dataset used to build local-population medical-simulation context. Designed so one completed record satisfies **US (HIPAA)**, **EU (GDPR)**, and **India (DPDP Act 2023)** at once. Field-level legend: **[SELF]** = self-attestable by a trained instructor; **[LEGAL]** = legal-review-required before the declaration can be relied on for that condition.

#### Section A — Identity & Provenance

| #  | Field                                     | What the user must state                                             | Notes / Regime hook                                                                                                     |
|----|-------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| A1 | <b>Declaration ID / version</b>           | Unique ID + version number                                           | Enables audit trail; re-declare on any material change                                                                  |
| A2 | <b>Data Steward</b>                       | Named individual + role + org + contact accountable for this dataset | GDPR "controller" locus; DPDP "Data Fiduciary" accountability (s.8); HIPAA workforce accountability                     |
| A3 | <b>Declaration date + review-due date</b> | Date signed; date for re-evaluation                                  | HIPAA Expert Determination and GDPR anonymisation are time-bound (re-identification risk drifts as external data grows) |

#### Section B — Data Nature & Source

| #  | Field                                   | What the user must state                                                                                                                                                                                                                | Notes / Regime hook                                                                                                                                                         |
|----|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B1 | <b>Data nature</b> (pick one)           | (a) <b>Factual — public</b> (already lawfully published), (b) <b>Factual — not-yet-public</b> (real but unpublished/embargoed/internal), (c) <b>Hypothetical / synthetic</b> (invented or generated, not derived from real individuals) | Drives which downstream fields are mandatory. (c) largely escapes all three regimes <b>if</b> genuinely non-derived; <b>[LEGAL]</b> if synthesized <i>from</i> real records |
| B2 | <b>Granularity</b> (pick one)           | (a) <b>Individual-level</b> (about identifiable persons), (b) <b>Aggregate / community-level</b> (counts, rates, prevalence by group/geo), (c) <b>Mixed</b>                                                                             | Individual-level triggers full de-id analysis; aggregate is often outside HIPAA PHI but still needs a stated basis (small cells can re-identify)                            |
| B3 | <b>Source &amp; type</b>                | Origin (registry, census/health-authority open data, EHR extract, literature, instructor knowledge, AI-generated), URL/citation or internal reference, and collection date                                                              | Establishes lawful origin; needed for GDPR "reasonably available information" and DPDP publicly-available analysis                                                          |
| B4 | <b>Population / geography described</b> | The cohort and smallest geographic unit represented                                                                                                                                                                                     | Small geo + small cohort = re-identification risk even in aggregates; feeds B2 and E-fields                                                                                 |

#### Section C — De-identification / Sanitization Method

| #  | Field                                            | What the user must state                                                                                                                                                                                                                                                                                                                                                                                      | Notes / Regime hook                                                                                         |
|----|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| C1 | <b>Method applied</b> (pick one+)                | (a) <b>HIPAA Safe Harbor</b> — all 18 identifiers removed, (b) <b>HIPAA Expert Determination</b> — qualified expert certified "very small" risk, (c) <b>GDPR-grade anonymisation</b> — irreversible; resists singling-out/linkability/inference, (d) <b>Pseudonymisation only</b> (key exists → still personal data), (e) <b>Not applicable</b> (data never contained personal data — B1=c or B2=b aggregate) | See mapping table below                                                                                     |
| C2 | <b>Safe Harbor checklist</b> (if C1=a)           | Confirm removal of each of the 18 identifiers (names; geo < state incl. ZIP unless first 3 digits of a >20,000-pop area; all date elements except year + age 90+ collapsed; phone; fax; email; SSN; MRN; health-plan #; account #; cert/license #; vehicle IDs; device IDs; URLs; IP; biometric; full-face photos; <b>any other unique identifying number, characteristic, or code</b> ) <b>[SELF]</b>        | 45 CFR 164.514(b)(2). Instructor can self-check the list                                                    |
| C3 | <b>Expert / statistical basis</b> (if C1=b or c) | Name/qualifications of expert, method, risk metric, and signed determination reference <b>[LEGAL]</b>                                                                                                                                                                                                                                                                                                         | 164.514(b)(1) requires a person with statistical expertise; GDPR anonymisation claim needs equivalent rigor |
| C4 | <b>Transformations performed</b>                 | Generalization, suppression, aggregation, noise/perturbation, k-anonymity/date-shifting etc.                                                                                                                                                                                                                                                                                                                  | Evidence the three GDPR attacks were addressed                                                              |

## Section D — Re-identification-Risk Statement

| #  | Field                               | What the user must state                                                                                                                                                          | Notes / Regime hook                                                                                |
|----|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| D1 | <b>Residual-risk assessment</b>     | Narrative: could a "motivated intruder"/anticipated recipient re-identify anyone, alone or combined with <b>reasonably available</b> other data? Rate very-small / low / residual | HIPAA "very small"; GDPR Recital 26 "means reasonably likely to be used"; WP29 05/2014 three risks |
| D2 | <b>Actual-knowledge attestation</b> | "I have no actual knowledge the retained data could identify an individual." <b>[SELF]</b> if C1=a; <b>[LEGAL]</b> otherwise                                                      | 164.514(b)(2)(ii) actual-knowledge clause                                                          |
| D3 | <b>Small-cell / aggregate check</b> | For aggregate data: confirm no cell is small enough to isolate an individual (e.g., minimum cell size / rare-attribute suppression) <b>[SELF]</b>                                 | Aggregate ≠ automatically safe                                                                     |

## Section E — Jurisdiction Attestations (all three, always)

| #  | Field                    | What the user must state                                                                                                                                                                                                                                                                                                                                                                                  | Legend                                                                                                     |
|----|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| E1 | <b>US / HIPAA</b>        | Either "Data is <b>outside HIPAA</b> because [no covered-entity/BA relationship OR never PHI OR aggregate non-PHI]" <b>[SELF]</b> or "Data <b>was PHI</b> and is de-identified via Safe Harbor <b>[SELF]</b> / Expert Determination <b>[LEGAL]</b> ."                                                                                                                                                     |                                                                                                            |
| E2 | <b>EU / GDPR</b>         | Either " <b>Anonymous</b> under Recital 26 (data protection principles do not apply)" — <b>[LEGAL]</b> to rely on — or " <b>Personal/pseudonymised data</b> ; lawful basis = [see F]." State that singling-out, linkability, inference were considered.                                                                                                                                                   | EDPB 04/2025 (anonymisation) & 01/2025 (pseudonymisation): pseudonymised data <b>remains</b> personal data |
| E3 | <b>India / DPDP 2023</b> | State one: " <b>Truly anonymised</b> → outside DPDP (identity cannot be inferred)" <b>[LEGAL]</b> ; or " <b>Personal data</b> ; ground = consent (s.4/6) or legitimate use (s.7)"; or " <b>Research/statistical exemption</b> (s.17(2)(b)) — no individual-level decisions + govt-prescribed safeguards." Note DPDP has <b>no sensitive-data tier and no statutory de-id safe harbor</b> . <b>[LEGAL]</b> | s.17 standards not yet fully notified (as of 2026) → conservative default                                  |

## Section F — Legal Basis / Consent (only if any of B1=a/b AND data is personal in any regime)

| #  | Field                                   | What the user must state                                                                      | Legend         |
|----|-----------------------------------------|-----------------------------------------------------------------------------------------------|----------------|
| F1 | <b>GDPR lawful basis</b>                | Art. 6 basis (+ Art. 9 condition if health data)                                              | <b>[LEGAL]</b> |
| F2 | <b>DPDP ground</b>                      | Consent + notice reference, or Section 7 legitimate use, or Section 17 research basis         | <b>[LEGAL]</b> |
| F3 | <b>Consent / authorization evidence</b> | Reference to consent record or HIPAA authorization, where the basis is consent                | <b>[LEGAL]</b> |
| F4 | <b>Not-yet-public data clearance</b>    | If B1=b: confirmation of rights/permission/embargo compliance to use unpublished factual data | <b>[LEGAL]</b> |

## Section G — Sign-off

| #  | Field                                      |                                                                                              |
|----|--------------------------------------------|----------------------------------------------------------------------------------------------|
| G1 | <b>Steward attestation &amp; signature</b> | "The above is accurate to my knowledge; I will re-declare on material change." <b>[SELF]</b> |
| G2 | <b>Legal review status</b>                 | Reviewed-by / date, <b>required</b> whenever any <b>[LEGAL]</b> field was invoked            |

## Method → Regime mapping (reference)

| Chosen method (C1)           | US / HIPAA                                               | EU / GDPR                                                                                | India / DPDP                                                |
|------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Safe Harbor (18 removed)     | De-identified, not PHI <b>[SELF]</b>                     | Strong, but <b>not automatically</b> GDPR-anonymous; assess residual risk <b>[LEGAL]</b> | Assess against "identity cannot be inferred" <b>[LEGAL]</b> |
| Expert Determination         | Not PHI if expert certifies "very small" <b>[LEGAL]</b>  | Often aligns with anonymisation, still assess 3 risks <b>[LEGAL]</b>                     | As above <b>[LEGAL]</b>                                     |
| GDPR-grade anonymisation     | Meets/exceeds HIPAA de-id                                | Outside GDPR (Recital 26) <b>[LEGAL]</b>                                                 | Likely outside DPDP <b>[LEGAL]</b>                          |
| Pseudonymisation only        | Still PHI (key exists)                                   | <b>Still personal data</b> (EDPB 01/2025)                                                | <b>Still personal data</b>                                  |
| N/A — aggregate/hypothetical | Likely outside HIPAA <b>[SELF]</b> , confirm small-cells | Outside GDPR if truly non-personal                                                       | Outside DPDP if identity not inferable                      |

**Self-attestable vs. legal-review-required — summary**

- **[SELF] (trained instructor):** data-nature/source classification (B1–B4), Safe Harbor 18-item checklist (C2), actual-knowledge for Safe Harbor (D2), small-cell aggregate check (D3), HIPAA "outside scope / Safe Harbor" attestation (E1), steward sign-off (G1).
- **[LEGAL] (counsel/DPO/qualified expert):** Expert Determination or any anonymisation *claim* (C3), GDPR Recital-26 "anonymous" reliance (E2), **all** DPDP scope/exemption conclusions (E3), every lawful-basis/consent field (F1–F4), and any use of not-yet-public factual data (F4). Rule of thumb: **claiming data is out-of-scope of a regime, or relying on a consent/legitimate-use basis, is [LEGAL]; merely stripping the HIPAA 18 identifiers is [SELF].**

# Appendix B — Provenance / source-declaration model

## Provenance / Source-Declaration Model

A single reusable object — the **SourceDeclaration** — attaches to (a) any individual **data element** and (b) any **community-block** (a curated group of elements: a locality profile, a prevalence layer, a scenario dataset). It is a thin projection of five standards, chosen so nothing here is invented: every field maps to PROV-O, DataCite 4.5, schema.org/Dataset, FORCE11, or C2PA.

**1. Core object — `SourceDeclaration` (per data element)**

| Field          | Type / enum            | Standard mapping                                                                                                         | Notes                                                                                                                                  |                                       |                                                                             |                                                                                    |
|----------------|------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| id             | URI / DOI / UUID       | DataCite Identifier ;<br>FORCE11 Unique Identification; PROV-O prov:Entity                                               | The provenance subject. Prefer a DOI; fall back to a persistent internal URI ( tbsim:elem/... ). Machine-actionable + globally unique. |                                       |                                                                             |                                                                                    |
| source_name    | string                 | DataCite Creator / Publisher ;<br>schema.org creator / publisher                                                         | Human-readable origin ("WHO Global Health Observatory", "SME panel 2026-Q2", "hypothetical for training").                             |                                       |                                                                             |                                                                                    |
| source_type    | enum: factual-public \ | factual-nonpublic \                                                                                                      | hypothetical \                                                                                                                         | expert-estimate                       | (project epistemic layer) → drives PROV-O relation + C2PA digitalSourceType | <b>The load-bearing field.</b> See §2. Answer "is this real, or can I show receipt |
| citation       | string (formatted)     | DataCite citation form: Creator (Year): Title. Publisher. ; schema.org citation ; FORCE11 Evidence                       | Rendered human citation. Auto-composed from the fields below.                                                                          |                                       |                                                                             |                                                                                    |
| url            | URL                    | schema.org url / sameAs ;<br>DataCite RelatedIdentifier                                                                  | Landing page or canonical location.                                                                                                    |                                       |                                                                             |                                                                                    |
| doi            | DOI string             | DataCite Identifier ;<br>FORCE11 Persistence                                                                             | Persistent id where one exists (empty allowed for non-public/hypothetical).                                                            |                                       |                                                                             |                                                                                    |
| retrieval_date | ISO-8601 date          | DataCite Date with dateType="Accessed" (use Collected for primary observation)                                           | When the value was pulled. Required for factual-public .                                                                               |                                       |                                                                             |                                                                                    |
| confidence     | enum: high \           | medium \                                                                                                                 | low                                                                                                                                    | (reuses FACE ENGINE confidence vocab) | Quality/certainty of the value. Pairs with is_estimate .                    |                                                                                    |
| is_estimate    | boolean                | (reuses FACE ENGINE is_estimate flag)                                                                                    | true ⇒ value is modeled/interpolated, not directly sourced. Auto-true for expert-estimate / hypothetical .                             |                                       |                                                                             |                                                                                    |
| quality_note   | string (opt)           | schema.org description                                                                                                   | Free-text caveat ("2019 data, extrapolated to 2026", "n<30").                                                                          |                                       |                                                                             |                                                                                    |
| steward        | string / ORCID / role  | PROV-O prov:wasAttributedTo + prov:Agent ; DataCite Contributor<br>role= DataManager ;<br>FORCE11 Credit and Attribution | Accountable owner of this declaration (person, ORCID, or role e.g. role:clinical-lead ).                                               |                                       |                                                                             |                                                                                    |
| derived_from   | list of id             | PROV-O prov:wasDerivedFrom / prov:hadPrimarySource ;<br>schema.org isBasedOn                                             | Lineage to parent element(s). Enables the chain FORCE11 Specificity & Verifiability requires.                                          |                                       |                                                                             |                                                                                    |

| Field            | Type / enum                        | Standard mapping                        | Notes                                              |
|------------------|------------------------------------|-----------------------------------------|----------------------------------------------------|
| license          | SPDX id / URL                      | schema.org license ;<br>DataCite Rights | Reuse terms of the source.                         |
| sanitization_ref | id of a<br>SanitizationDeclaration | (project)                               | Cross-link to the sibling<br>declaration — see §5. |

## 2. source\_type — the epistemic enum (and how it drives the standards)

| source_type       | Meaning                                                                                 | PROV-O relation emitted                                               | C2PA digitalSourceType (IPTC)                            | Citation obligation                                                                     |
|-------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------|
| factual-public    | Verifiable, publicly reachable fact (public dataset, stats agency, peer-reviewed)       | prov:hadPrimarySource                                                 | .../digitalSourceType/dataDrivenMedia                    | <b>MUST</b> have url or doi + retrieval_date                                            |
| factual-nonpublic | Real but access-restricted (licensed feed, institutional record, private communication) | prov:wasDerivedFrom +<br>prov:wasAttributedTo                         | .../dataDrivenMedia                                      | MUST have steward + access note; doi optional                                           |
| expert-estimate   | SME judgement / Delphi / calibrated guess                                               | prov:wasAttributedTo<br>(Agent = the expert/panel)                    | .../humanEdits (or custom<br>com.tbsim.expertEstimate )  | MUST name the expert/panel in source_name + set is_estimate=true                        |
| hypothetical      | Invented for training; no real-world referent                                           | (no source relation) +<br>com.tbsim.synthetic:<br>{real_person:false} | .../trainedAlgorithmicMedia or<br>com.tbsim.hypothetical | is_estimate=true , confidence reflects plausibility only; <b>must never claim a doi</b> |

This mirrors the FACE ENGINE, which already marks synthetic content with `com.tbsim.synthetic:{synthetic:true, real_person:false}` and an IPTC `digitalSourceType . hypothetical` here is the data-side twin of that image-side marking.

## 3. Community-block wrapper — ProvenanceManifest (per community-block)

A community-block gets one manifest that (a) carries block-level DataCite mandatory metadata and (b) **rolls up** the per-element declarations.

| Field                                       | Standard mapping                                                                                                             |                       |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| identifier (DOI/URI)                        | DataCite Identifier (M)                                                                                                      |                       |
| title                                       | DataCite Title (M); schema.org name                                                                                          |                       |
| creators[] , publisher ,<br>publicationYear | DataCite Creator / Publisher / PublicationYear (M)                                                                           |                       |
| resourceTypeGeneral = Dataset               | DataCite ResourceType (M)                                                                                                    |                       |
| elements[] →<br>SourceDeclaration.id        | schema.org hasPart ; each element is a prov:Entity                                                                           |                       |
| provenance_summary                          | counts by source_type (e.g. {factual-public: 12, expert-estimate: 3, hypothetical: 5}) — the block's "receipt at a glance"   |                       |
| weakest_link                                | min(confidence) + list of un-cited factual-* elements → the block cannot pass export while any factual-* element lacks `url` | doi + retrieval_date` |
| c2pa_manifest                               | the signed manifest (§4)                                                                                                     |                       |
| sanitization_manifest_ref                   | link to the block's sanitization roll-up (§5)                                                                                |                       |

## 4. Machine-checkable reference-list export (two serializations, one source of truth)

(a) **schema.org/Dataset + JSON-LD** — web-discoverable, human-citable. Each element becomes an entry; lineage uses `isBasedOn` ; sources use `citation` ; the block uses `Dataset` with `distribution` , `license` , `creator` , `identifier` . This is the form Google Dataset Search / FAIR tooling consumes.



```
{ "@context":"https://schema.org/", "@type":"Dataset",
 "name":"East-Region locality prevalence block v3",
 "identifier":"https://doi.org/10.xxxx/tbsim.block.eastreg.v3",
 "creator":[{"@type":"Organization","name":"TBSIM Service Community"}],
 "license":"https://spdx.org/licenses/CC-BY-4.0",
 "hasPart":[
 {"@type":"Dataset","name":"malaria seasonal incidence",
 "citation":"WHO GH0 (2024): Malaria incidence. WHO.",
 "sameAs":"https://www.who.int/data/gho",
 "isBasedOn":"tbsim:elem/who-gho-malaria-2024",
 "additionalProperty":[
 {"@type":"PropertyValue","name":"source_type","value":"factual-public"},
 {"@type":"PropertyValue","name":"retrieval_date","value":"2026-06-30"},
 {"@type":"PropertyValue","name":"confidence","value":"high"},
 {"@type":"PropertyValue","name":"steward","value":"role:clinical-lead"}}]
]
}
```

**(b) PROV-O / PROV-JSON graph** — the verifiable lineage graph. Every `SourceDeclaration` → `prov:Entity`; `steward` → `prov:Agent` via `wasAttributedTo`; `derived_from` → `wasDerivedFrom` / `hadPrimarySource`; `retrieval_date` → a `prov:Activity` (`prov:used` the external source, `prov:endedAtTime` = `retrieval_date`). This is what satisfies FORCE11 *Specificity & Verifiability*.

**Validator (the "if asked, prove it" gate)** — export MUST fail if any of:

- an element with `source_type` ∈ `{factual-public}` has no `url|doi` or no `retrieval_date` (FORCE11 *Evidence + Access*),
- an element with `source_type=hypothetical` carries a `doi` (false-provenance guard),
- any element lacks a `steward` (FORCE11 *Credit/Attribution*),
- any `derived_from` id is dangling (broken lineage),
- the block lacks the 6 DataCite mandatory properties.

Pass ⇒ emit both serializations + the C2PA-bound copy.

**(c) C2PA carrier** — when the block or element is exported *as/with an asset*, the same facts ride inside a `com.tbsim.provenance` custom assertion in the C2PA manifest, alongside the FACE ENGINE's existing `c2pa.actions` and `com.tbsim.synthetic` assertions, with source elements listed as C2PA ingredients ( `relationship:"inputTo"` ). This makes provenance cryptographically bound and tamper-evident, not just adjacent metadata.

5. Composition with the sanitization declaration

They are **orthogonal, co-located, and mutually referential** — two assertions in one manifest answering two different questions:

|           | Sanitization declaration                                                                                                                                                | Provenance / source declaration                                                                      |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Question  | "Is this element <b>safe to expose</b> ?" (PII stripped, k-anon, synthetic-substituted)                                                                                 | " <b>Where did this element come from</b> and can I cite it?"                                        |
| Verb      | <i>transforms/removes</i>                                                                                                                                               | <i>attributes/traces</i>                                                                             |
| PROV view | it is a <code>prov:Activity</code> ( <code>san:sanitize</code> ) that <code>prov:used</code> the raw element and <code>prov:wasGeneratedBy</code> → the exposed element | the <code>prov:Entity</code> + its <code>wasDerivedFrom</code> / <code>hadPrimarySource</code> edges |
| C2PA      | <code>com.tbsim.sanitization</code> assertion + <code>c2pa.actions:{action:"c2pa.redacted"}</code>                                                                      | <code>com.tbsim.provenance</code> assertion + <code>ingredients</code>                               |

How they compose:

1. Each `SourceDeclaration.sanitization_ref` points to the `SanitizationDeclaration` for the same element (and vice-versa), so a reviewer can pivot from "cite it" to "is it clean" in one hop.
2. In PROV-O they chain naturally: `raw_source` --`wasDerivedFrom`--> `sanitized_element` , where the sanitization Activity sits on that edge ( `used` raw, `generated` sanitized, `wasAssociatedWith` the steward). Provenance describes the *ends* of the edge; sanitization describes the *edge*.
3. The reference-list export includes a `sanitization_status` badge per element ( `raw` / `sanitized` / `synthetic-substitute` ) pulled from the linked sanitization declaration — so one export shows **both** the receipt *and* the safety stamp.
4. Both roll up into the same community-block manifest and the same signed C2PA manifest store, so a single verification pass proves *source + safety + no-tamper* together. Critically, when sanitization replaces a real value with a synthetic one, the element's `source_type` flips to `hypothetical` / `expert-estimate` and the provenance chain records the substitution — the two declarations stay consistent by construction.

Appendix C — Community-block exchange format (interop detail)

Community-Block Exchange Format v1.0.0

A portable, aggregate-only JSON document describing a real clinical service community, consumed by V8 + V9 to localize simulations and by FACE GENERATOR to build a demographically-matched face cohort. **Invariant: no patient-level rows, ever.** Everything is population-level statistics, public/aggregate sources, and synthetic targets.

Design decisions (standard → where it lands)

| Concern                       | Standard aligned to                                                                         | How it lands in the block                                                                                                                                                                                                                                    |
|-------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geography                     | GeoJSON RFC 7946                                                                            | <code>geography.boundary</code> is a raw GeoJSON <code>Feature / FeatureCollection</code> ; WGS84 only (CRS84 implicit — RFC 7946 forbids a <code>crs</code> member); <code>bbox</code> allowed                                                              |
| Demographic mix + prevalence  | FHIR R4 <b>MeasureReport</b> stratifier/stratum + <b>Group.characteristic</b>               | strata carry <code>code + value</code> (CodeableConcept-shaped: <code>system / code / display</code> ) and a <code>fraction</code> (0–1) or <code>count</code> ; mirrors summary-MeasureReport <code>stratum.population.count</code> without listing members |
| Condition coding              | OMOP CDM vocab pattern / FHIR CodeableConcept                                               | each condition binds <code>code.system</code> (SNOMED/ICD-10/OMOP <code>concept_id</code> ) + <code>code.code</code> + human <code>display</code> ; <code>prevalence</code> as rate per 100k or fraction                                                     |
| Wrapper metadata              | schema.org/Dataset                                                                          | <code>meta</code> mirrors <code>spatialCoverage</code> , <code>temporalCoverage</code> , <code>variableMeasured</code> , <code>license</code> , <code>creator</code> , <code>distribution</code>                                                             |
| Integrity / signing           | RFC 8785 (JCS) + JWS                                                                        | <code>integrity</code> block: canonicalize sans- <code>integrity</code> , SHA-256 → <code>content_hash</code> , detached JWS → <code>signature</code>                                                                                                        |
| Population strata definitions | US Census / CDC BRFSS / WHO (FACE ENGINE <code>population_stratification_schema.md</code> ) | <code>sex · age bands · race/ethnicity</code> axes reused verbatim so the block round-trips with GPM                                                                                                                                                         |

```

{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "$id": "https://tbsim.company/schemas/community-block/1.0.0.json",
 "title": "CommunityBlock",
 "type": "object",
 "required": ["format", "spec_version", "identity", "geography", "demographics", "sanitization", "provenance"],
 "additionalProperties": false,
 "properties": {

 "format": { "const": "tbsim.community-block" },
 "spec_version": { "type": "string", "pattern": "^\\d+\\.\\d+\\.\\d+$",
 "description": "SemVer of THIS schema. Consumer routes on major." },

 "identity": {
 "type": "object",
 "required": ["block_id", "block_version", "name", "created_at", "created_by"],
 "additionalProperties": false,
 "properties": {
 "block_id": { "type": "string", "format": "uuid",
 "description": "Stable across versions of the same community." },
 "block_version": { "type": "string", "pattern": "^\\d+\\.\\d+\\.\\d+$",
 "description": "SemVer of the CONTENT (data revisions), independent of spec_version." },
 "name": { "type": "string", "maxLength": 200 },
 "description": { "type": "string", "maxLength": 2000 },
 "created_at": { "type": "string", "format": "date-time" },
 "created_by": { "type": "string", "description": "Instructor/org display name – no personal email/PII." },
 "supersedes": { "type": ["string", "null"], "format": "uuid",
 "description": "block_id this revision replaces, if any." },
 "tags": { "type": "array", "items": { "type": "string" } }
 }
 },

 "geography": {
 "type": "object",
 "required": ["scope", "boundary"],
 "additionalProperties": false,
 "properties": {
 "scope": { "enum": ["nation", "state", "county", "tract", "zcta", "hsa", "custom"] },
 "label": { "type": "string", "description": "Human place name, e.g. 'Cook County, IL'." },
 "gazetteer": { "type": "object",
 "description": "Optional external IDs: FIPS, GEOID, ISO-3166, OSM relation.",
 "additionalProperties": { "type": "string" } },
 "boundary": { "$ref": "#/$defs/GeoJSON",
 "description": "RFC 7946 Feature or FeatureCollection. WGS84 (CRS84) ONLY; no `crs` member." },
 "bbox": { "type": "array", "items": { "type": "number" }, "minItems": 4, "maxItems": 6 }
 }
 },

 "demographics": {
 "type": "object",
 "required": ["population_estimate", "strata"],
 "additionalProperties": false,
 "properties": {
 "population_estimate": { "type": "integer", "minimum": 0,
 "description": "Aggregate head-count estimate. NOT a member list (FHIR Group.quantity semantics)." },
 "as_of": { "type": "string", "description": "ISO-8601 temporalCoverage of the estimate." },
 "strata": { "type": "array", "items": { "$ref": "#/$defs/Stratum" },
 "description": "One entry per demographic axis value; FHIR MeasureReport.stratifier.stratum shape." }
 }
 },

 "clinical_prevalence": {
 "type": "array",

```

```

"description": "Locally prevalent conditions → drives case/scenario weighting.",
"items": {
 "type": "object",
 "required": ["condition", "measure"],
 "additionalProperties": false,
 "properties": {
 "condition": { "$ref": "#/$defs/CodeableConcept" },
 "measure": {
 "type": "object",
 "required": ["type", "value"],
 "additionalProperties": false,
 "properties": {
 "type": { "enum": ["prevalence_fraction", "rate_per_100k", "incidence_per_100k"] },
 "value": { "type": "number", "minimum": 0 },
 "ci_low": { "type": ["number", "null"] },
 "ci_high": { "type": ["number", "null"] }
 }
 },
 "stratified_by": { "type": "array", "items": { "$ref": "#/$defs/Stratum" },
 "description": "Optional demographic breakdown of THIS condition." },
 "source_ref": { "type": "array", "items": { "type": "string" },
 "description": "IDs into provenance.sources[]." }
 }
}
},

"sdo": {
 "type": "array",
 "description": "Social determinants – aggregate indices only (e.g. SVI, ADI, uninsured %, food-desert %).",
 "items": {
 "type": "object",
 "required": ["factor", "value"],
 "additionalProperties": false,
 "properties": {
 "factor": { "$ref": "#/$defs/CodeableConcept",
 "description": "e.g. system:'gravity-sdo' or 'cdc-svi', code:'SVI_OVERALL'." },
 "value": { "type": "number" },
 "unit": { "type": "string", "description": "e.g. 'percentile', 'fraction', 'index'." },
 "source_ref": { "type": "array", "items": { "type": "string" } }
 }
 }
},

"face_cohort_spec": {
 "type": "object",
 "description": "Demographic targets handed to FACE GENERATOR (/api/geo/generate).",
 "additionalProperties": false,
 "properties": {
 "cohort_size": { "type": "integer", "minimum": 1, "maximum": 2000 },
 "targets": {
 "type": "array",
 "description": "Target fractions per demographic cell; SUM of fraction across a given axis ≈ 1.0.",
 "items": {
 "type": "object",
 "required": ["axis", "value", "fraction"],
 "additionalProperties": false,
 "properties": {
 "axis": { "enum": ["sex", "age_band", "race_ethnicity"] },
 "value": { "type": "string", "description": "Axis value from FACE ENGINE M1 matrix." },
 "fraction": { "type": "number", "minimum": 0, "maximum": 1 }
 }
 }
 }
 }
},

```

```

 "derive_from_demographics": { "type": "boolean", "default": true,
 "description": "If true, FACE GENERATOR may compute targets from `demographics.strata` and ignore explicit `targets`." },
 "synthetic_only": { "const": true,
 "description": "Guardrail: faces are AI-generated + labeled; block asserts no real portraits." }
 }
},

"sanitization": {
 "type": "object",
 "description": "Machine-checkable 'contains no PII' declaration. REQUIRED. Import MUST reject if contains_pii=true.",
 "required": ["contains_pii", "aggregate_only", "min_cell_size", "method", "declared_at", "declared_by"],
 "additionalProperties": false,
 "properties": {
 "contains_pii": { "const": false,
 "description": "Hard invariant. Any value but false → import MUST fail closed." },
 "aggregate_only": { "const": true,
 "description": "Asserts no patient-level rows; all figures are population statistics." },
 "min_cell_size": { "type": "integer", "minimum": 1, "default": 11,
 "description": "Small-cell suppression threshold applied (HIPAA-style k≥11 recommended)." },
 "suppressed": { "type": "boolean", "default": true,
 "description": "Cells below min_cell_size were suppressed/rounded." },
 "method": { "type": "string",
 "description": "e.g. 'small-cell-suppression+rounding', 'public-aggregate-passthrough'." },
 "synthetic_faces": { "type": "boolean", "default": true },
 "declared_at": { "type": "string", "format": "date-time" },
 "declared_by": { "type": "string", "description": "Role/org, not personal identity." }
 }
},

"provenance": {
 "type": "object",
 "description": "schema.org/Dataset-style source list. Every stat SHOULD reference a source id.",
 "required": ["sources"],
 "additionalProperties": false,
 "properties": {
 "sources": {
 "type": "array", "minItems": 1,
 "items": {
 "type": "object",
 "required": ["id", "name", "access"],
 "additionalProperties": false,
 "properties": {
 "id": { "type": "string", "description": "Local ref id, e.g. 'src-census-acs-2024'." },
 "name": { "type": "string" },
 "publisher": { "type": "string", "description": "e.g. 'US Census Bureau', 'CDC BRFSS', 'WHO'." },
 "url": { "type": ["string", "null"], "format": "uri" },
 "vintage": { "type": ["string", "null"], "description": "Data year/period (temporalCoverage)." },
 "license": { "type": ["string", "null"], "description": "SPDX id or free text; public-domain expected." },
 "access": { "enum": ["public", "aggregate-public", "synthetic", "derived"] },
 "retrieved_at": { "type": ["string", "null"], "format": "date-time" }
 }
 }
 }
 },
 "generator": { "type": "string",
 "description": "Tool + version that built the block, e.g. 'gpm@1.4.0 / face-engine@0.9.0'." }
}
},

"meta": {
 "type": "object",
 "description": "schema.org/Dataset projection for discovery/catalog. Optional, non-authoritative.",
 "additionalProperties": true,
 "properties": {

```

```

 "license": { "type": "string" },
 "spatialCoverage": { "type": "string" },
 "temporalCoverage": { "type": "string", "description": "ISO-8601 interval." },
 "variableMeasured": { "type": "array", "items": { "type": "string" } },
 "keywords": { "type": "array", "items": { "type": "string" } }
 }
},

"integrity": {
 "type": "object",
 "description": "Computed over the block with `integrity` REMOVED, canonicalized per RFC 8785 (JCS).",
 "required": ["canonicalization", "hash_alg", "content_hash"],
 "additionalProperties": false,
 "properties": {
 "canonicalization": { "const": "RFC8785" },
 "hash_alg": { "enum": ["SHA-256", "SHA-384", "SHA-512"] },
 "content_hash": { "type": "string", "pattern": "^[a-f0-9]{64,128}$",
 "description": "hex digest of JCS(block \\ integrity)." },
 "signature": {
 "type": ["object", "null"],
 "additionalProperties": false,
 "description": "Optional detached JWS over the same canonical bytes.",
 "properties": {
 "alg": { "enum": ["EdDSA", "ES256", "RS256"] },
 "kid": { "type": "string", "description": "Key id / cert thumbprint." },
 "jws": { "type": "string", "description": "Detached JWS (RFC 7515) – payload omitted." }
 }
 }
 }
}
},

"$defs": {

 "CodeableConcept": {
 "type": "object",
 "required": ["system", "code"],
 "additionalProperties": false,
 "description": "FHIR-shaped coding. system=vocabulary URI/OID, code=concept id, display=human label.",
 "properties": {
 "system": { "type": "string", "description": "e.g. 'http://snomed.info/sct', 'icd-10-cm', 'omop-concept'." },
 "code": { "type": "string" },
 "display": { "type": "string" }
 }
 },

 "Stratum": {
 "type": "object",
 "required": ["axis", "value"],
 "additionalProperties": false,
 "description": "FHIR MeasureReport.stratifier.stratum shape. Provide fraction OR count (fraction preferred).",
 "properties": {
 "axis": { "type": "string", "description": "e.g. 'sex', 'age_band', 'race_ethnicity', 'payer'." },
 "value": { "$ref": "#/$defs/CodeableConcept" },
 "fraction": { "type": ["number", "null"], "minimum": 0, "maximum": 1 },
 "count": { "type": ["integer", "null"], "minimum": 0,
 "description": "Aggregate count. MUST be ≥ sanitization.min_cell_size or null (suppressed)." }
 }
 },

 "GeoJSON": {
 "type": "object",
 "description": "RFC 7946 object. type ∈ Feature|FeatureCollection. No `crs` member (WGS84/CRS84 implicit).",

```

```

 "required": ["type"],
 "properties": {
 "type": { "enum": ["Feature", "FeatureCollection"] }
 },
 "not": { "required": ["crs"] }
 }
}
}

```

## Export / import discipline

### SemVer (two independent axes).

- `spec_version` — the schema itself. Consumers route on **major**: accept same-major, reject unknown-major, tolerate unknown-minor/patch (see forward-compat).
- `identity.block_version` — the *content* revision of a community; `identity.block_id` is stable across content revisions, `supersedes` chains lineage.

### Forward-compat rules.

- Additive-only within a major: new **optional** fields → minor bump; clarifications/fixes → patch. Removing/renameing a field or tightening a required set → **major** bump.
- Consumers MUST **ignore unknown fields** at the top level and inside `meta / gazetteer` (those are `additionalProperties: true`). Everything else is `additionalProperties: false` — so unknown keys there are a hard reject, which is what forces a major bump for structural change.
- **Fail closed on major mismatch; warn-and-proceed on minor ahead; accept on minor behind.**

### Signing / hashing (RFC 8785 + JWS).

1. Build the full block including all data, `integrity` absent (or stripped).
2. `canonical = JCS(block)` per RFC 8785 (sorted keys UTF-16, normalized numbers, no whitespace).
3. `content_hash = hex(SHA-256(canonical))` → write into `integrity`.
4. Optional: detached JWS over the **same** `canonical` bytes → `integrity.signature.jws`.
5. Verify by stripping `integrity`, re-canonicalizing, recomputing the digest, comparing, then verifying the JWS. This is the standard JCS "remove signature field → canonicalize → sign/verify" pattern, so it's identical in Python and TS.

### "Contains no PII" invariant (enforced, not advisory).

- `sanitization.contains_pii` is a `const false`; `aggregate_only` is `const true`; `face_cohort_spec.synthetic_only` and `sanitization.synthetic_faces` assert synthetic portraits. An importer **MUST** reject any block where these fail — schema validation alone enforces the invariant because the constants can't be anything else and still validate.
- Structural guardrail: the format has **no field capable of holding a patient-level row** — no member arrays, no identifiers beyond org/role display strings, no free-text narrative on individuals. PII can't be represented, so it can't leak. Small-cell suppression (`min_cell_size`, default  $k \geq 11$ ) protects re-identification via tiny strata; suppressed cells carry `count: null`.

### Import algorithm (both V8 and V9).

1. JSON-parse → validate against community-block/<major>.json
2. Check `spec_version` major == supported (else reject)
3. Recompute JCS hash; compare to `integrity.content_hash` (else reject: tampered)
4. If `integrity.signature` present: verify JWS against trusted kid (else reject/quarantine)
5. Assert `sanitization.contains_pii === false` && `aggregate_only === true` (else reject)
6. Upsert by `identity.block_id`; if `block_version` <= existing, no-op unless `supersedes` chain says otherwise
7. Project: geography-GPM, demographics+clinical\_prevalence-case weighting + local-context, `face_cohort_spec`→FACE GENERATOR /api/geo/generate

## Python (pydantic v2) + TS parity

The schema is intentionally flat, closed (`additionalProperties:false` except two catalog maps), and uses only JSON primitives + enums, so it generates cleanly both ways.



```
pydantic v2 – abbreviated; full model mirrors the schema 1:1
from typing import Literal, Optional
from pydantic import BaseModel, Field, ConfigDict

class CodeableConcept(BaseModel):
 model_config = ConfigDict(extra="forbid")
 system: str
 code: str
 display: Optional[str] = None

class Sanitization(BaseModel):
 model_config = ConfigDict(extra="forbid")
 contains_pii: Literal[False] # invariant: only False validates
 aggregate_only: Literal[True]
 min_cell_size: int = Field(default=11, ge=1)
 method: str
 declared_at: str
 declared_by: str

class CommunityBlock(BaseModel):
 model_config = ConfigDict(extra="forbid")
 format: Literal["tbsim.community-block"]
 spec_version: str = Field(pattern=r"^\d+\.\d+\.\d+$")
 identity: "Identity"
 geography: "Geography"
 demographics: "Demographics"
 sanitization: Sanitization
 provenance: "Provenance"
 clinical_prevalence: list["ClinicalPrevalence"] = []
 sdoh: list["Sdoh"] = []
 face_cohort_spec: Optional["FaceCohortSpec"] = None
 meta: Optional[dict] = None
 integrity: Optional["Integrity"] = None
```

```
// TS – same contract; validate with Ajv against the JSON Schema at runtime
export interface CommunityBlock {
 format: "tbsim.community-block";
 spec_version: `${number}.${number}.${number}`;
 identity: Identity;
 geography: Geography;
 demographics: Demographics;
 clinical_prevalence?: ClinicalPrevalence[];
 sdoh?: Sdoh[];
 face_cohort_spec?: FaceCohortSpec;
 sanitization: Sanitization; // contains_pii:false & aggregate_only:true are literal types
 provenance: Provenance;
 meta?: Record<string, unknown>;
 integrity?: Integrity;
}

export interface Sanitization {
 contains_pii: false; aggregate_only: true;
 min_cell_size: number; method: string; declared_at: string; declared_by: string;
}
```

**Canonicalize/hash helpers:** Python `rfc8785` (pip) → `hashlib.sha256`; TS `canonicalize` (npm, RFC 8785) → `WebCrypto/crypto.subtle.digest('SHA-256')`. Both emit identical bytes, so a block signed in V9-cloud verifies byte-identically when imported into V8-on-prem.

## Appendix D — Population & geospatial data-sources catalog

# Population & Geospatial Health Data for Medical-Simulation Context

A practical briefing on authoritative prevalence sources, synthetic-population methods, place-to-profile pipelines, and responsible gap-filling — the "ready reference base" the Service Community module points to.

## 1. Authoritative sources by region

United States — all US federal data is public domain (17 U.S.C. §105), freely reusable commercially with attribution

| Source                            | Region                                        | What it provides                                                                                                                          | Access / API                                                | License                                 |
|-----------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Census Bureau / ACS               | US, to <b>block group</b>                     | Demographic mix: age, sex, race/ethnicity, income, education, housing, insurance                                                          | REST API (free key); ACS 5-Year 2009–2024                   | Public domain                           |
| CDC PLACES                        | US: county, place, <b>census tract</b> , ZCTA | Model-based <b>local prevalence</b> , 40 measures (12 outcomes, 7 preventive, 4 risk behaviors, 7 disabilities, 3 status, 7 social needs) | Socrata API / bulk (PLACES Data Portal); 2025 tract release | Public domain                           |
| BRFSS                             | US: national, state, <b>SMART</b> metro       | The telephone-survey behavioral/risk data PLACES is modeled from; direct state & metro prevalence                                         | Annual files, WEAT, SMART                                   | Public domain                           |
| CDC WONDER                        | US: national→county                           | Mortality, natality, cancer, infectious disease, population counts (outcomes/mortality, not point-prevalence)                             | Menu query + WONDER API                                     | Public domain                           |
| County Health Rankings & Roadmaps | US: <b>county</b>                             | Ranked composite: health behaviors, chronic conditions, outcomes, access, SDOH                                                            | CSV downloads; maps                                         | Free reuse w/ attribution (check Terms) |
| AHRQ SDOH Database                | US: county, ZIP, <b>tract</b> , 2009–2020     | Curated SDOH from 44 sources across 5 domains, pre-linked by geography                                                                    | Bulk CSV/Excel                                              | Public domain                           |
| HRSA Area Health Resources Files  | US: <b>county</b>                             | Health workforce, facilities, population characteristics, utilization — supply-side/access                                                | HRSA Data Warehouse download                                | Public domain                           |

## EU / Global

| Source                                   | Region                                    | What it provides                                                         | Access / API                                                                        | License                                                                                       |
|------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Eurostat                                 | EU + EEA, NUTS levels                     | Health status, causes of death, healthcare + demographics/socioeconomics | Free SDMX 3.0 & JSON APIs (no key)                                                  | Free reuse (commercial) w/ attribution                                                        |
| WHO Global Health Observatory (GHO)      | Global, 194 states (mostly country-level) | 1,000+ health indicators                                                 | OData API, no auth (⚠ current OData API deprecating ~end 2025 — plan for successor) | Open/free (verify per dataset)                                                                |
| IHME Global Burden of Disease (GBD 2023) | Global → many countries subnational       | Deep harmonized prevalence/incidence/DALYs by cause, age, sex, year      | GBD Results Tool + GHDx                                                             | ⚠ <b>Non-commercial free; commercial use requires a paid University of Washington license</b> |

## India

| Source                           | Region                                          | What it provides                                                                     | Access / API                                                     | License                                 |
|----------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------|
| NFHS-5 (2019–21)                 | India: national, state/UT, <b>707 districts</b> | Population, health, nutrition; diabetes/BP, anemia, maternal/child, disability, WASH | Factsheets (OGD India); microdata (World Bank Microdata Library) | Open (OGD India NDSAP; WB registration) |
| Census of India                  | India, to village/ward                          | Demographic denominators, household amenities                                        | Portal bulk tables                                               | Government open data                    |
| NSSO / NSO health rounds (MoSPI) | India, national + state                         | Socio-economic + health expenditure/morbidity rounds                                 | MoSPI microdata + reports                                        | Government open data                    |

## 2. Synthetic-population methods

- Synthea** — open-source generator of full **synthetic patient records** (FHIR/C-CDA/CSV); seeds from US Census demographics to city level, drives disease progression via 90+ guideline-based clinical modules. *Good for:* longitudinal, clinically-coherent individual EHRs for a locale. *Limits:* US-centric defaults; localizing to a non-US/fine-grained place needs demographics + geo files swapped and module frequencies tuned. Validated against clinical quality measures.
- Iterative Proportional Fitting (IPF) / synthetic reconstruction** — fits a seed microdata sample to known **marginal totals** (census cross-tabs) so the synthetic population's joint distribution matches the area's margins. *Good for:* a demographically-representative population mix when you only have aggregate margins. Refinements: Hierarchical IPF, relative-entropy minimization, combinatorial optimization. A ready US national synthetic-populations dataset exists as a shortcut.

- **Composition:** IPF answers "*who lives here*" (denominator); prevalence sources (PLACES/GBD/NFHS) answer "*what conditions do they carry*"; Synthea turns each synthetic person into a coherent record. Order: reconstruct population → attach condition probabilities by stratum → instantiate records.

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### 3. Place → profile (repeatable, provenance attached at every step)

1. **Resolve geography** — place → canonical geo-ID (US tract/ZCTA/FIPS; EU NUTS; India district): the downstream join key.
2. **Demographic denominator** — ACS (US) / Eurostat (EU) / Census+NFHS (India) → age×sex×race/ethnicity×income×education×insurance for that geo-ID (record source, vintage, geo level).
3. **Condition & SDOH prevalence** — join CDC PLACES (US tract), AHRQ SDOH, County Health Rankings; GBD/GHO (global); NFHS district (India). Store each measure with estimate, source, year, resolution.
4. **Reconstruct population** (IPF against ACS/census margins), then **assign conditions** by drawing on stratum-specific prevalence (not a flat area average) — preserves within-area heterogeneity.
5. **(Optional) Instantiate records** (Synthea or equivalent) for full clinical detail.
6. **Emit a provenance manifest** — every field → {value, source, source\_url, vintage, geo\_level, method}.

**Resolution caveat:** prevalence sources are model-based small-area (PLACES) or survey (BRFSS/NFHS) estimates with uncertainty; carry the geo level and flag when a coarser-geography value is borrowed for a finer place.

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### 4. Gap-filling — responsible representation of missing data

Never silently invent; **label the epistemic status**. Two cases:

- **(a) Factual-but-not-yet-integrated** — real & knowable but not yet in the pipeline (newer release, local registry). Tag `status: sourced-pending` + citation, confidence **High**; treat as authoritative once verified.
- **(b) Hypothetical / expert-estimate** — no public figure exists. Tag `status: estimated`, record the **method** (interpolated from neighboring geography / analogous population / clinician judgment), the **basis**, and a confidence band (**Medium/Low**). Never present with the same weight as sourced values.

**Per-value field schema:**

```
{ value, unit,
 status: sourced | sourced-pending | estimated,
 confidence: high | medium | low,
 source, source_url, vintage, geo_level,
 method, // "PLACES tract estimate" | "IPF from ACS margins" | "SME estimate 2026"
 note } // rationale/caveat, surfaced in the sim UI
```

Build rules: confidence is first-class (sim renders "estimated" distinctly; instructors can filter to sourced-only for high-stakes debriefs); prefer sourced-with-borrowed-resolution over invented; estimates keep method+author so they're auditable and replaceable; **commercial-license watch** — prefer US federal + Eurostat; gate GBD behind the licensing decision.

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### 5. Two load-bearing licensing facts (commercial product)

1. **US federal sources (ACS, PLACES, BRFSS, WONDER, AHRQ, HRSA) are public domain / freely reusable commercially**; Eurostat permits commercial reuse with attribution.
2. **IHME GBD is free only for non-commercial use — commercial use requires a paid University of Washington license**. Also budget for the WHO GHO OData API successor (current one deprecating ~end 2025).